Delta DPS-1125AB 1450W Reverse Engineering

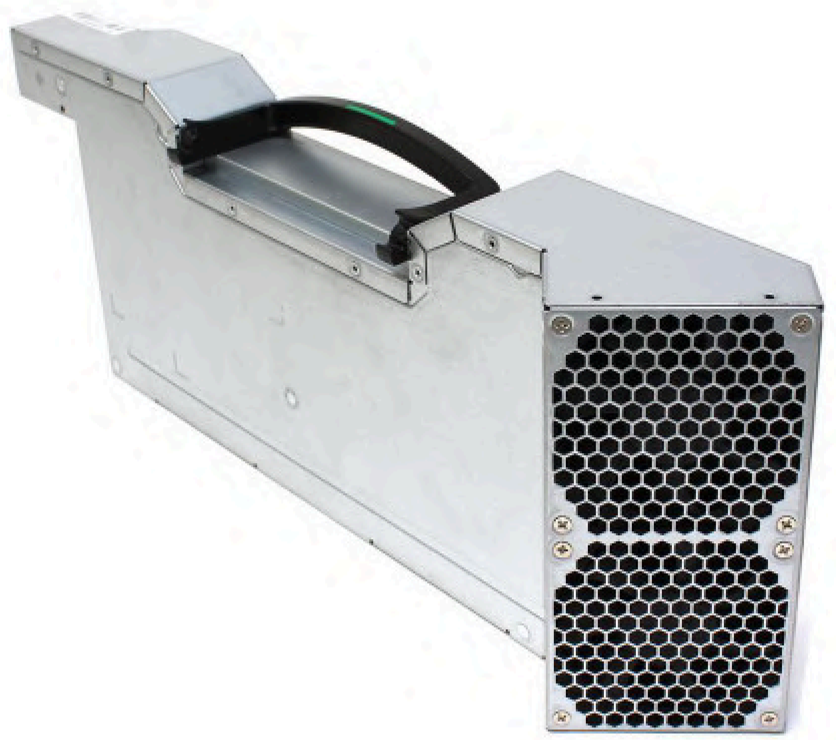

Table of contents

Overview

The signaletic plate and the outside view

Figure 1

Figure 2

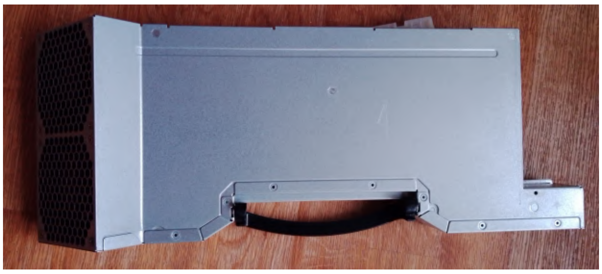

Figure 3

The PCB

Dimensions :

- Thickness ~1.6mm
- Length ~ 306mm
- max-width ~ 162mm
- middle-width ~ 128mm

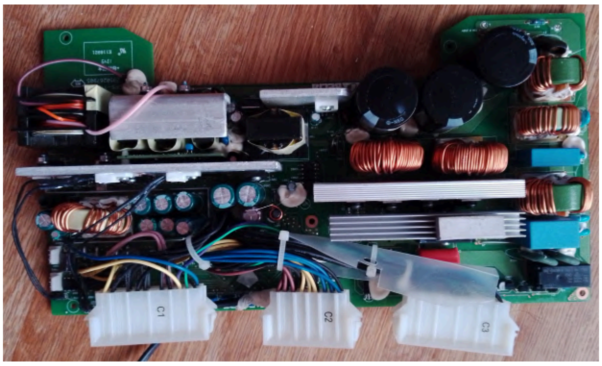

Figure 4

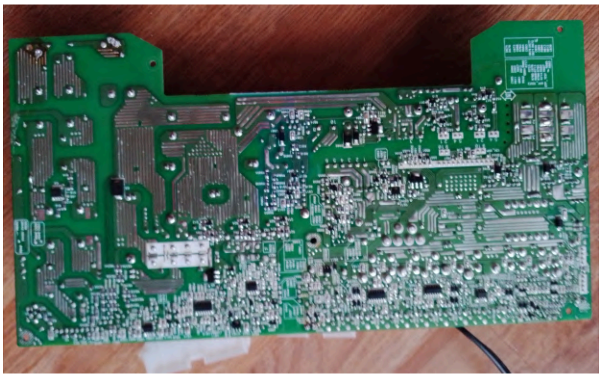

Figure 5

Power supply schematic

Below is the schematic of some principal parts of the power supply. The purpose is to illustrate the main components rather than provide a 100% accurate schematic.

Block Diagram of the Schematic

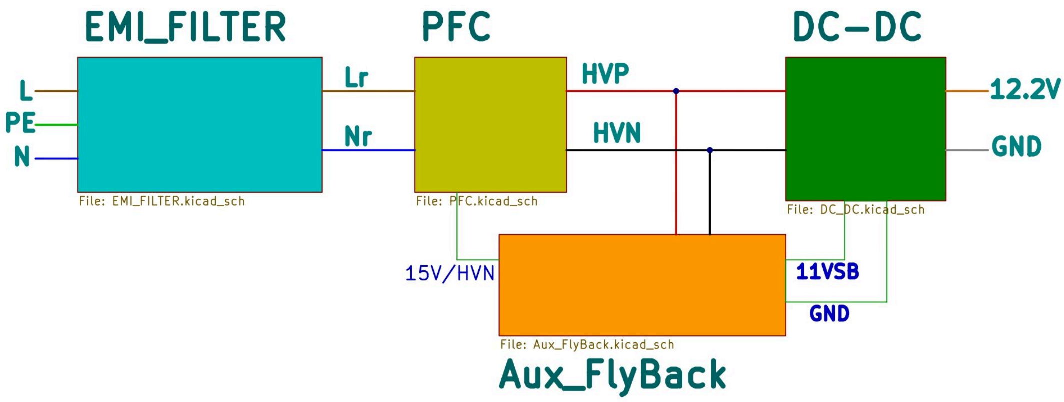

Figure 6

The EMI Inptut filter

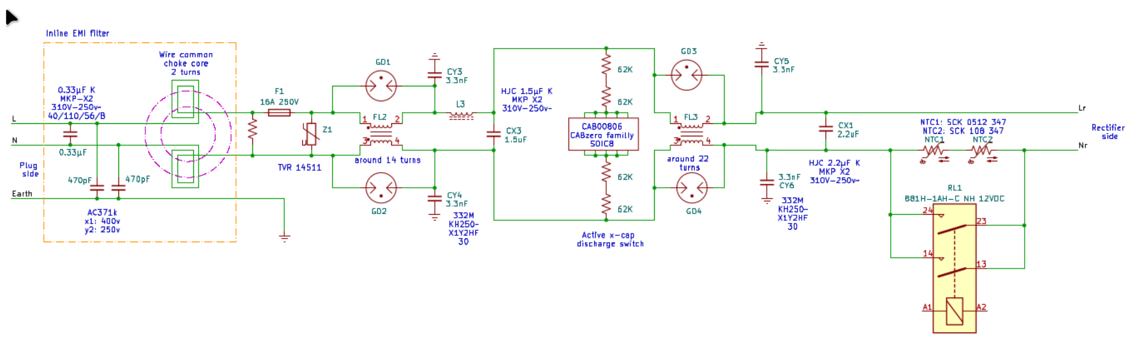

Figure 7

The emi filter has a active discharge switch. The idea is to put a small circuit; when the grid voltage is applied, the switch is open. When the user removes the plug, the circuit detects the absence of the 50–60 Hz voltage variation and closes the circuit, so the X-capacitors can be discharged. The discharge resistors are optimized to work only when the plug is removed (small resistor size and simpler thermal management).

The power factor correction (PFC)

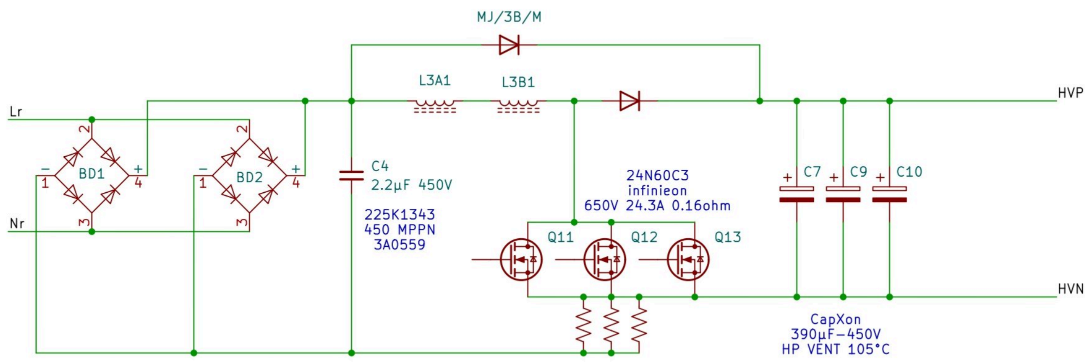

Figure 8

The PFC is a classical hard-switching PFC using 3 MOSFETs and 3 electrolytic capacitors.

The controller is apparently CM6502S (marking CM6502SX) see datasheet. The output driver is built around the NPN transistor PB4350.

The Flyback Auxiliary Circuit

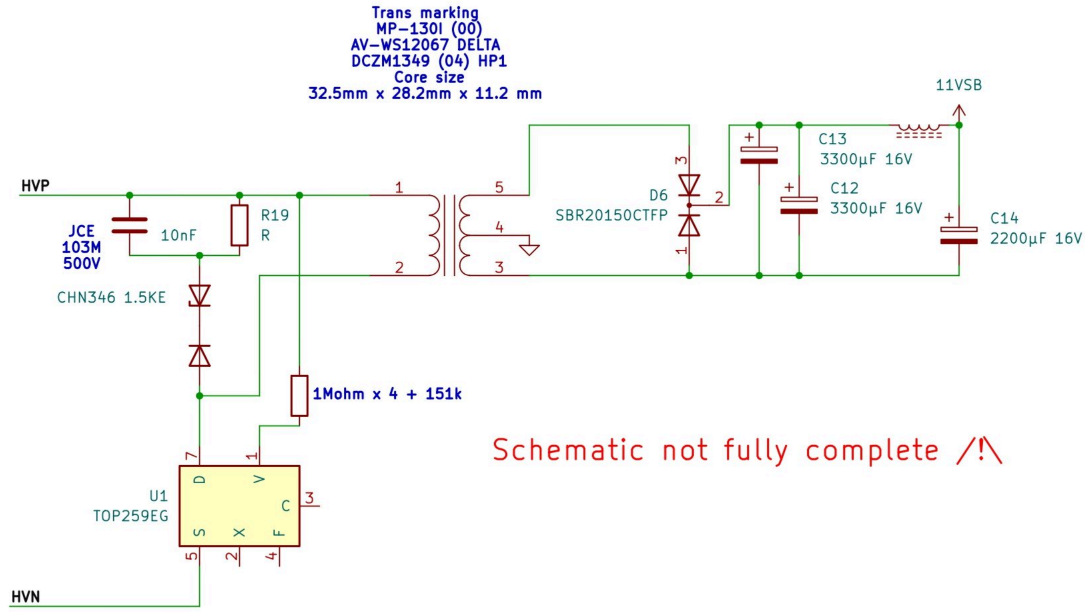

Figure 9

The auxiliary power supply is built around the TOP259EG controller.

The Main Isolated DC-DC

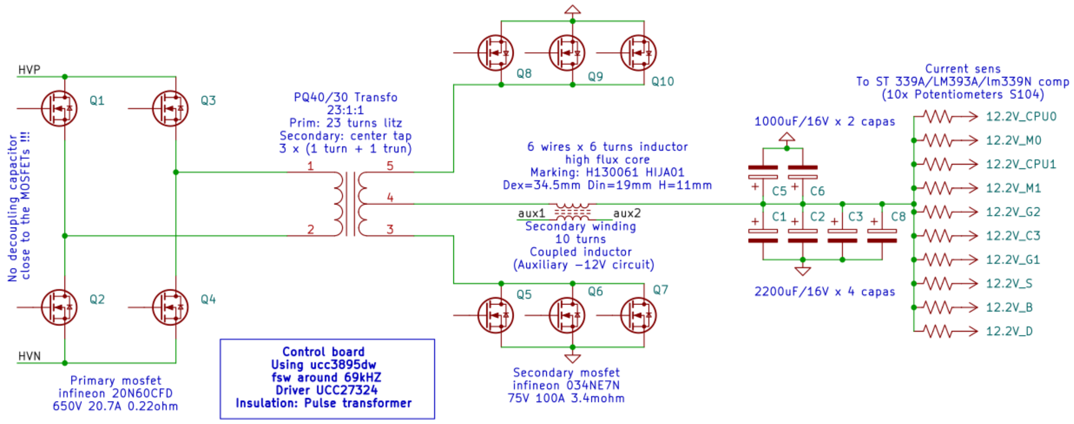

Figure 10

The isolated 400V-12V DC-DC is a phase shifted full bridge (**PSFB**) converter.

- The main transformer is apparently a no gapped core PQ40/30.
- Primary mosfet is infinient mosfet 20N60CFD 650V 20.7A 220mΩ
- Secondary mosfet is infinient mosfet 034NE7N 75V 100A 3.4mΩ
- The output inductor is composed of 6 parallel wires (D= 1.2mm), 6 turns, plus a small wire with 10 turns(D= 0.8mm) for auxilary -12V output: see below
- The controller is ucc3895dw, and driver is ucc27324

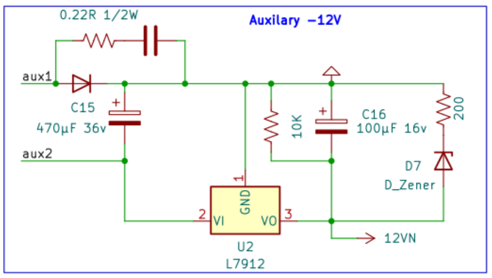

Figure 11

Component analysis

The main transformer (DC-DC transformer)

Overview of the transformer

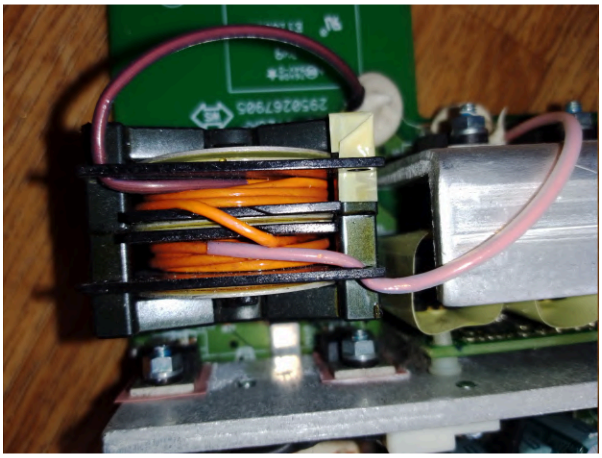

Figure 12

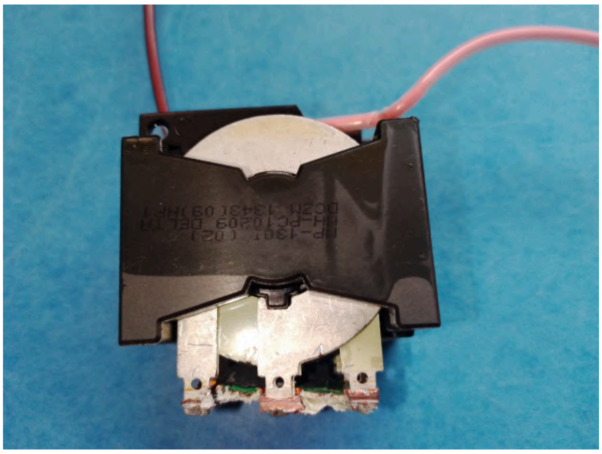

Figure 13

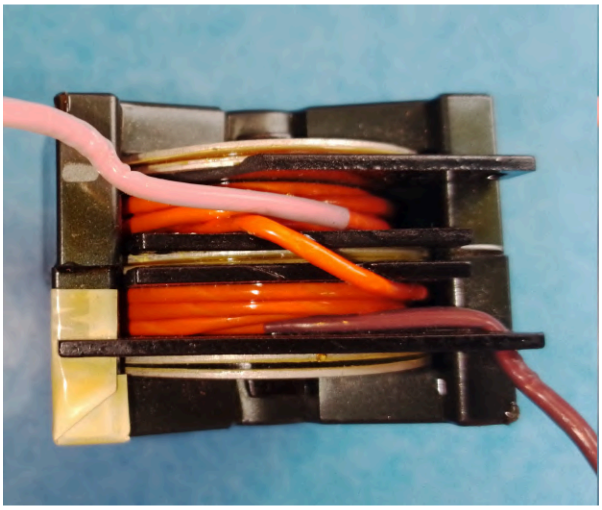

Figure 14

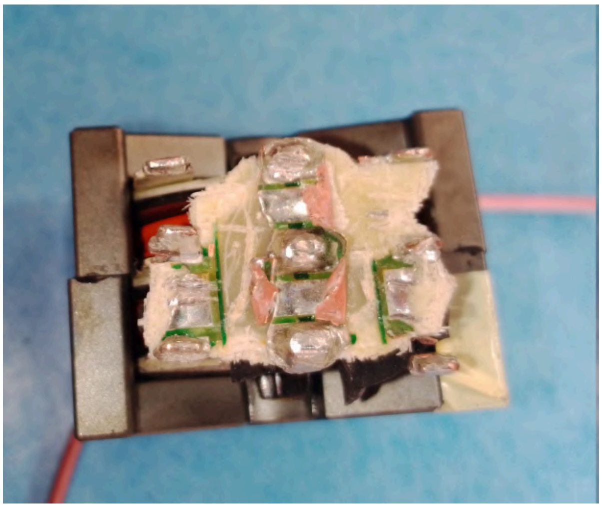

Figure 15

Figure 16 The transformer apparently has no air gap

Dimension of the transformer

Below the dimensions of PQ40/30, see the datasheet.

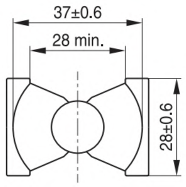

Figure 17

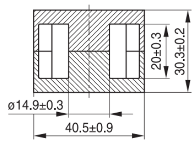

Figure 18

Datasheet measurement of PQ40/30 (mm)- Nominal	Actual core dimensions (mm)
40.5	40
30.3	30
28	28.2 and 27.8
14.9	14.9 ad 15
5	5

Regarding the dimensions and the shape of the transformer, it appears to be a TDK PQ40/30 core.

Characterization of the inductance factor (AL)

A small signal test is used to cacacterize the AL of the core.

- Setup: The function generator (50 Ω) powers the primary winding (23 turns). The input voltage and input current are measured

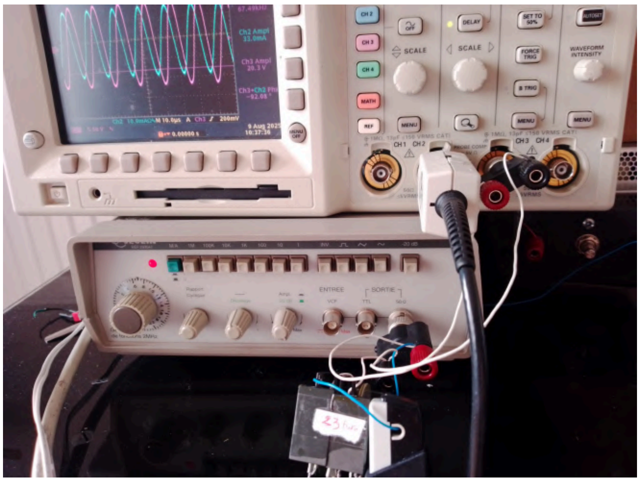

Figure 19

- Scope screenshot:

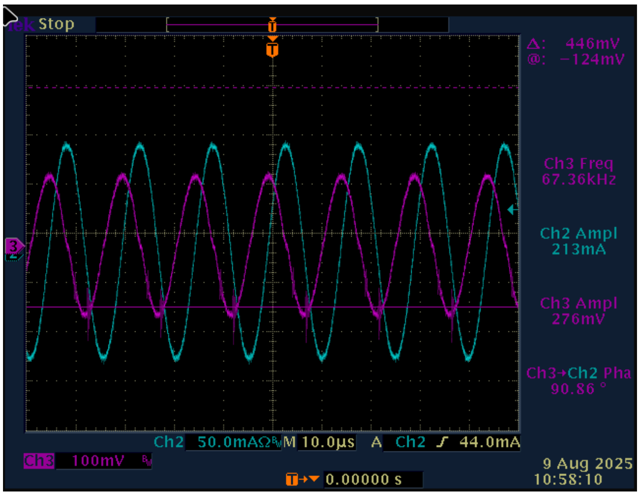

Figure 20

- Result (5 shots):
  - Measurement

	Freqkhz	ch2_mA	ch3_v	phase_deg
0	67.10	33.0	20.2	-91.75
1	67.25	33.2	20.2	-91.06
2	67.67	33.2	20.3	-93.63
3	67.32	33.0	20.2	-95.06
4	67.49	33.0	20.3	-92.08

```
- Calculating the inductance using voltage, current and frequency
```

	Freqkhz	ch2_mA	ch3_v	phase_deg	L_uH
0	67.10	33.0	20.2	-91.75	1451.894436
1	67.25	33.2	20.2	-91.06	1439.929167
2	67.67	33.2	20.3	-93.63	1438.076235
3	67.32	33.0	20.2	-95.06	1447.149683
4	67.49	33.0	20.3	-92.08	1450.650532

```
- The average value is:
```

$$\mathrm{Lm} = 1446 \ \ (\mu\mathrm{H})$$

```
- The inductance factor
```

$$A_L = \frac{\mathrm{Lm} \cdot 1 \times 10^3}{(N)^2} = \frac{1446 \cdot 1 \times 10^3}{(23)^2} \qquad = 2733.459 \ \ (\mathrm{nH/turns}\ 2)$$

From TDK datasheet of PQ40/30 cores the closed reference to this inductance factor is N92 or N49 with 3900 nH.

From Ferroxcube datasheet, the closed values are 3F4 (2500nH)and 3F36 (3500 nH).

If we assume that this core corresponds to one of these 2 datasheet cores, we can assume that the actual core is a PQ40/30 3F4.

$$A_{Lnom} = 2500 \ \ (\mathrm{nH})$$
$$\mathrm{error} = 100 \cdot \frac{A_L - A_{Lnom}}{A_{Lnom}} = 100 \cdot \frac{2733.459 - 2500}{2500} \qquad = 9.338 \ \ (\backslash\%)$$

The litz wire

Number of wires	84
External diameter (mm), copper + insulation	1.5mm
Wire diameter (mm)	0.1mm
Total length	176cm

Figure 21

Figure 22

Secondary high-current conductors

Figure 23 the shape of one of secondary conductor

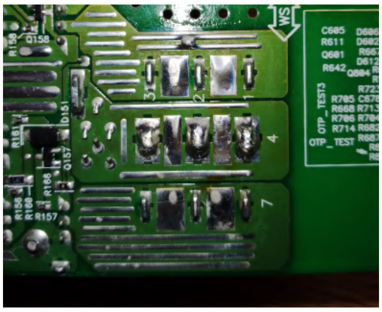

Figure 24 The secondary: PCB side

The secondary turns are apparently tin-plated copper. When cut, the copper color is visible.

Dimensions: width =8.5mm , thickness =0.8/0.9mm

The secondary consists of 6 loops. Each pair of loops forms a center-tapped 1:1 secondary, and the three secondaries are connected in parallel for high-current operation.

The winding arrangement is **SS-PP-SS-PP-SS** (see bellow)
This arrangement is known for its effectiveness in reducing leakage inductance.

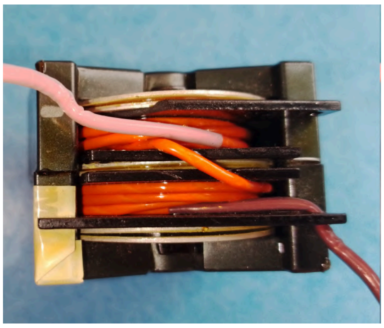

Figure 25 SS-PP-SS-PP-SS arrangement

Estimating the Peak AC Flux Density (B_{peak})

n a Phase-Shifted Full-Bridge (PSFB), Full-Bridge primary combined with a Center-Tapped secondary:

$$V_{out} = V_{in} \cdot \left(\frac{N_s}{N_p}\right) \cdot D_{eff}$$

Where:

- V_{in} = Input DC link voltage.
- V_{out} = Output DC voltage.
- N_p = Number of turns on the primary winding.

- N_s = Number of turns on *one half* of the center-tapped secondary winding (since you specified $n:1:1$, your ratio $\frac{N_s}{N_p} = \frac{1}{n}$).
- D_{eff} = effective duty cycle of the primary switches, maximum **1.0 (100%)**:

$$D_{eff} = \frac{\phi}{\pi} = \frac{\phi^\circ}{180^\circ}$$

Because your transformer is specified as $n:1:1$, the primary has n turns, and each half of the secondary has 1 turn.

$\frac{N_s}{N_p} = \frac{1}{n}$ so:

$$V_{out} = \frac{V_{in} \cdot D_{eff}}{n}$$

And

$$D_{eff} = \frac{V_{out} \cdot n}{V_{in}}$$

Real-World Adjustments

To get an accurate calculation for a real circuit:

$$V_{out} = \left(V_{in} \cdot \frac{1}{n} \cdot D_{eff}\right) - V_{drop}$$

- V_{drop} **:** Voltage drop of the secondary synchronous MOSFETs, voltage reduction caused by the dead-time, duty cycle loss from leakage inductance, and other drops in copper wires.

- Application using the ideal formula:

$$\begin{aligned} V_{out} &= 12.200 \ \ (\mathrm{V}) \\ V_{in} &= 400 \ \ (\mathrm{V}) \\ n &= 23 \ \ (\mathrm{turns}) \\ D_{eff} &= \frac{V_{out} \cdot n}{V_{in}} = \frac{12.200 \cdot 23}{400} \qquad = 0.701 \end{aligned}$$

let's add 5% or margin (to cover the voltage drops)

$$D_{eff} = 0.737$$

- The formula of B_{peak}

Full Bridge Max Phase Shift ($\phi = 180^\circ$, $D_{eff} = 1$)

$$V_{in} = 4 \cdot B_{\text{peak}} \cdot A_e \cdot N_p \cdot f_s$$

so

$$B_{\text{peak}} = \frac{V_{in}}{4 \cdot A_e \cdot N_p \cdot f_s}$$

For Other Phase Shifts ($\phi < 180^\circ$)

$$V_{in} = \frac{4 \cdot B_{\text{peak}} \cdot A_e \cdot N_p \cdot f_s}{D_{eff}}$$

so

$$B_{\text{peak}} = \frac{V_{in} \cdot D_{eff}}{4 \cdot A_e \cdot N_p \cdot f_s}$$

$$\begin{aligned}
f_s &= \frac{135 \times 10^3}{2} & &= 67500.000 \\
\text{Aemin}_{mm} &= 174.400 \quad (\text{mm2 see PQ40/30 datasheet}) \\
\text{Np} &= 23 \\
\text{Ae}_{min} &= \text{Aemin}_{mm} \cdot 1 \times 10^{-6} = 174.400 \cdot 1 \times 10^{-6} & &= 0.000 \ (\\
B_{peak} &= 1 \times 10^3 \cdot V_{in} \cdot \frac{D_{eff}}{4 \cdot \text{Ae}_{min} \cdot \text{Np} \cdot f_s} = 1 \times 10^3 \cdot 400 \cdot \frac{0.737}{4 \cdot 0.000 \cdot 23 \cdot 67500.000} & &= 272.044 \ (
\end{aligned}$$

From Ferroxcube datasheet, Tthe $Bsat_{min}$ of 3F4 is ***330mT*** (@ 100°C & 25kHz)

Assuming our hypothesis and calculation : operating at ~82% of the saturation limit is risky.

The output inductor (output filter LC)

Figure 26 The removed inducotr

Figure 27 The test winding

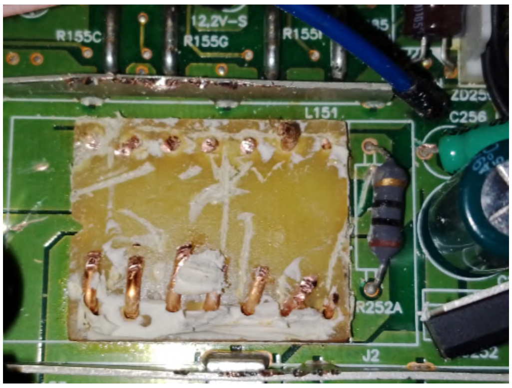

Figure 28

Dimensions:

- D-ext = 33.3 mm
- D-int = 19.2 mm
- H = 10.8 mm
- Width (D-ext - D-int) = 6.9/7 mm

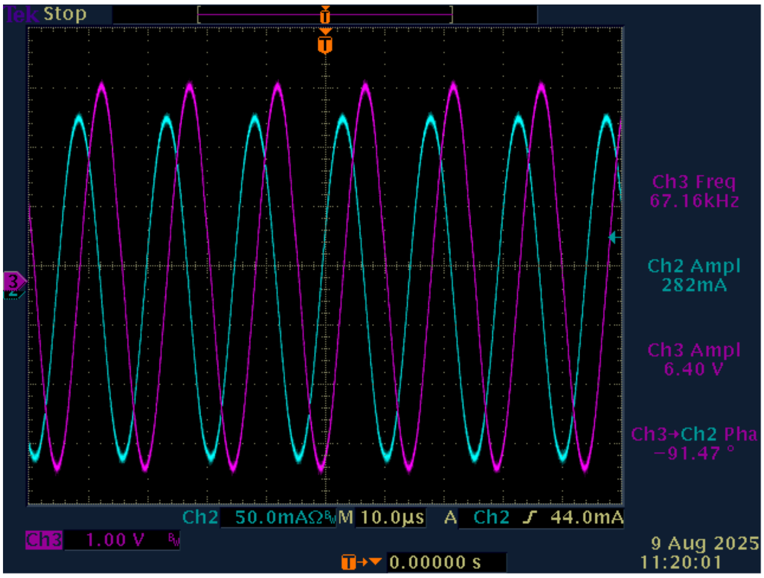

Figure 29 Small signal characterization (28 turns)

$$
\begin{aligned}
N &= 28 \ \ (\text{turns}) \\
V &= 6400 \ \ (\text{mv}) \\
I &= 282 \ \ (\text{mA}) \\
f &= 67160.000 \ \ (\text{Hz}) \\
L_{uH} &= 1 \times 10^6 \cdot \frac{V}{I \cdot f \cdot 2 \cdot \pi} = 1 \times 10^6 \cdot \frac{6400}{282 \cdot 67160.000 \cdot 2 \cdot 3.142} && = 53.782 \ \ (\mu\text{H}) \\
\text{AL} &= 1000 \cdot \frac{L_{uH}}{(N)^2} = 1000 \cdot \frac{53.782}{(28)^2} && = 68.600 \ \ (\text{nH})
\end{aligned}
$$

Regarding the dimesions, the inductance factor,this core is apparently a high-flux core.

The closed core that i found in internet is C058324A2, see the datasheet.

The control board

The controller

The controller is ucc3895dw, and driver is ucc27324

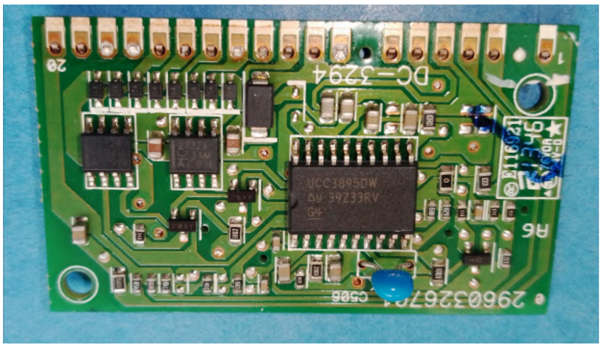

Figure 30

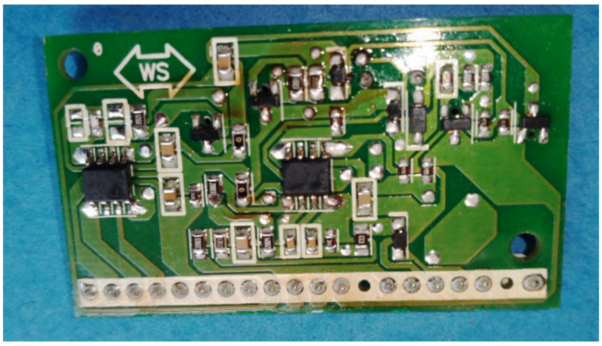

Figure 31

- The CT capacitor (the blue one) marking is 680pF (code 681)
- The RT resistor is 100kΩ (code 1003) from the controller ucc3895dw datasheet , the switching frequency formula:

$$
\begin{aligned}
\mathrm{CT} &= 6.800 \times 10^{-10} \quad (\mathrm{F}) \\
\mathrm{RT} &= 1.000 \times 10^{5} \quad (\Omega) \\
\mathrm{Fosc} &= \frac{1}{\mathrm{RT} \cdot \mathrm{CT}} = \frac{1}{1.000 \times 10^{5} \cdot 6.800 \times 10^{-10}} \qquad = 1.471 \times 10^{4}
\end{aligned}
$$

- t_{OSC} Formula (see ucc3895dw datasheet , page 14)

$$t_{OSC} = \frac{5 \times R_T \times C_T}{48} + 120\,\mathrm{ns}$$

where:

- C_T is in Farads
- R_T is in Ohms
- t_{OSC} is in seconds
- C_T can range from 100 pF to 880 pF

$$
\begin{aligned}
t_{OSC} &= \frac{5 \cdot \mathrm{RT} \cdot \mathrm{CT}}{48} + 120 \times 10^{-9} = \frac{5 \cdot 1.000 \times 10^{5} \cdot 6.800 \times 10^{-10}}{48} + 120 \times 10^{-9} \qquad = 7.203 \times 10^{-6} \\
f_{OSC} &= \frac{1}{t_{OSC}} = \frac{1}{7.203 \times 10^{-6}} \qquad = 1.388 \times 10^{5} \\
f_{sw} &= \frac{f_{OSC}}{2} = \frac{1.388 \times 10^{5}}{2} \qquad = 6.941 \times 10^{4} \\
f_{swkHz} &= 1 \times 10^{-3} \cdot f_{sw} = 1 \times 10^{-3} \cdot 6.941 \times 10^{4} \qquad = 6.941 \times 10^{1}
\end{aligned}
$$

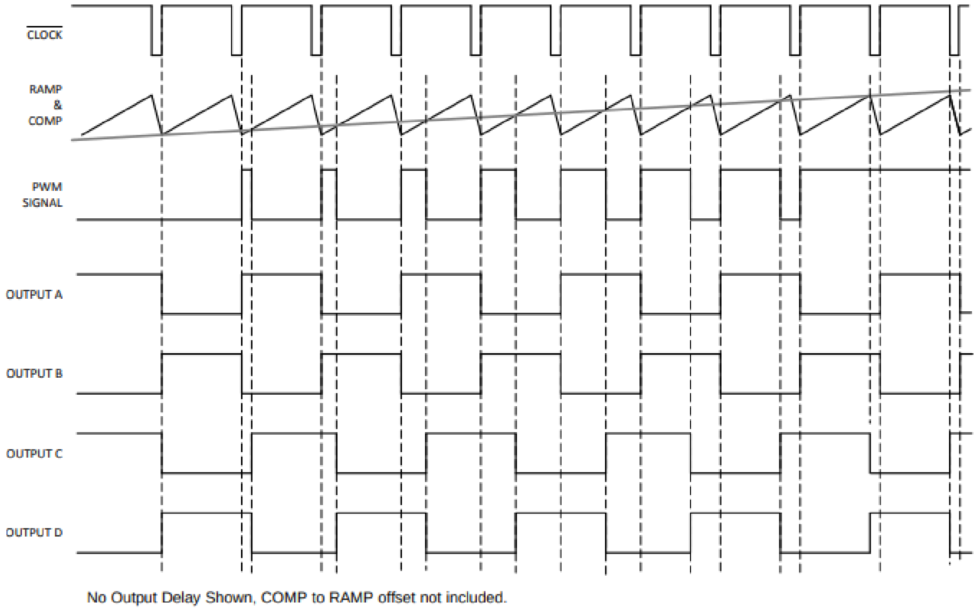

Figure 16. UCC3895 Timing Diagram

Figure 32 datasheet timing

Below the measurement of CT voltage, we isolate the control board, power it with 12V (Vcc, GND).

Therorical value of f_{OSC} is 138.8kHZ, measurement 135kHz.

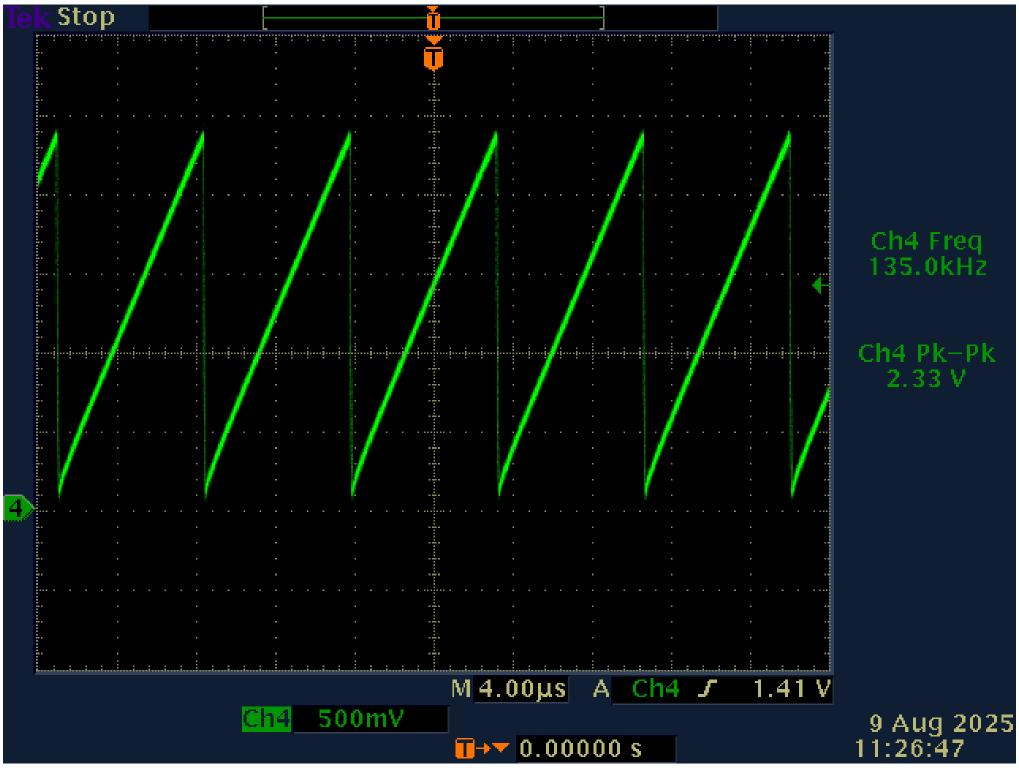

Figure 33 CT capacitor voltage

The driver insulation transformers

Since the control board use 2 no isolated drivers ucc27324, so the designer shose to use gate driver transformer to control the mosfets.

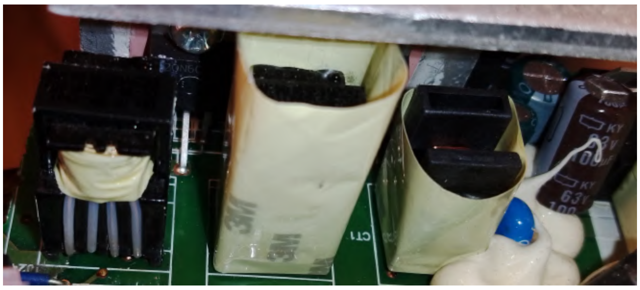

Figure 34

The fans

In this power supplay we found 2 DELTA fans QUR0812sh DC12V 0.50A

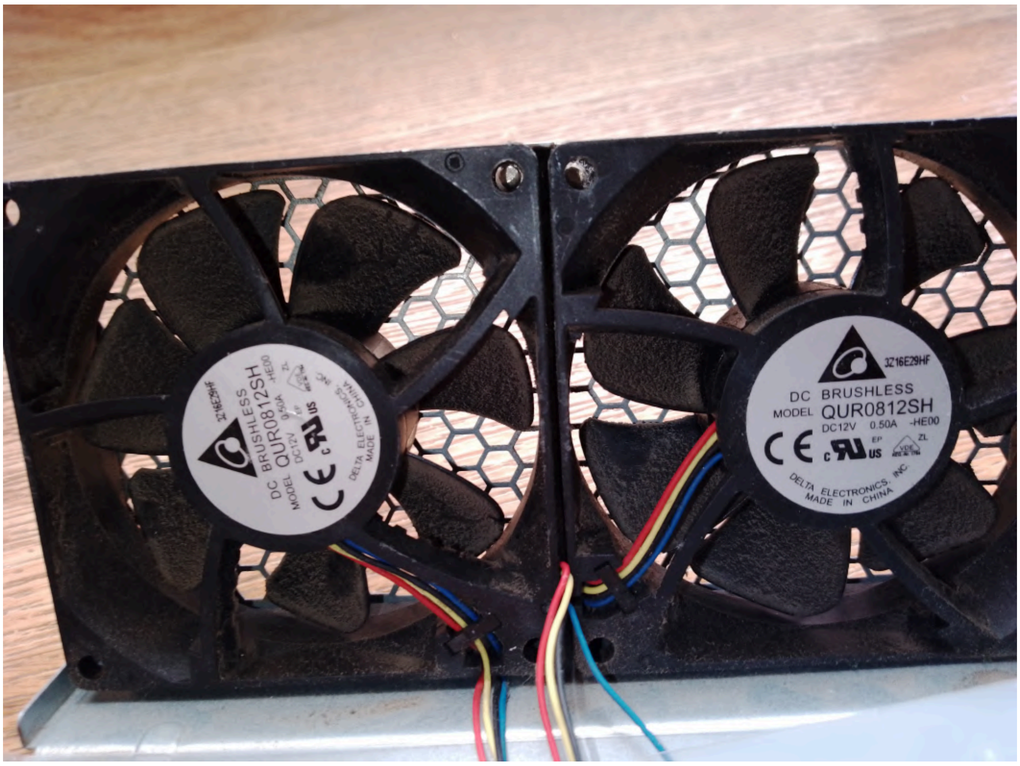

Figure 35 QUR0812sh DC12V 0.50A

From this fan datasheet, this fan has a PWM wire to control the speed.

The inline EMI filter

For more information, please check the schematics above

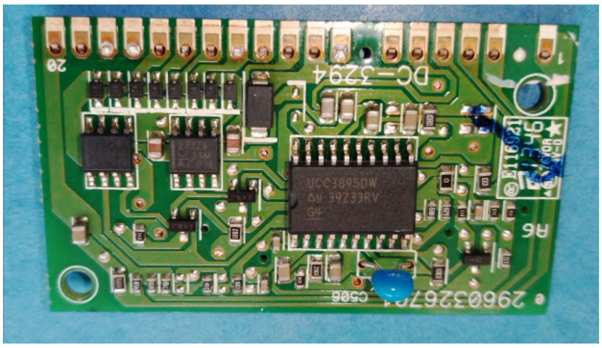

Figure 36

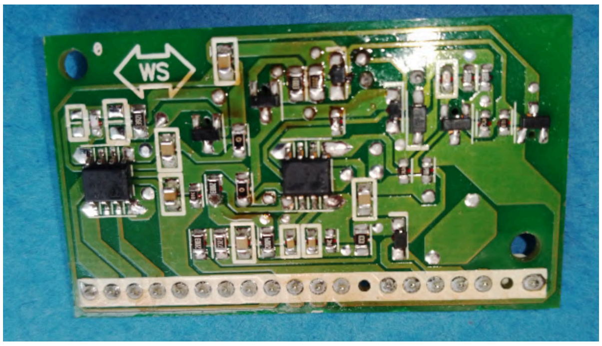

Figure 37

Figure 38

Figure 39

Figure 40

The housing

Below some images of the housing, we didn't check if the insulation is a high temperature plastic or normal polymeter

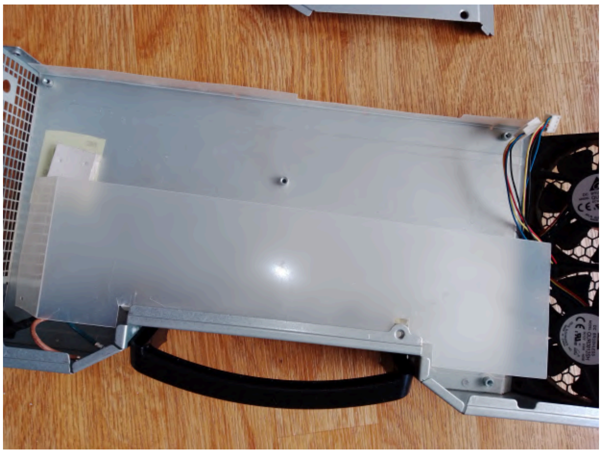

Figure 41

Figure 42

Figure 43

Figure 44

The flyback circuit

For more information, please check the schematics above.

We didn't turn ON the power supply, after many check we found the the flyback circuit (TOP259EG) is dead, see below:

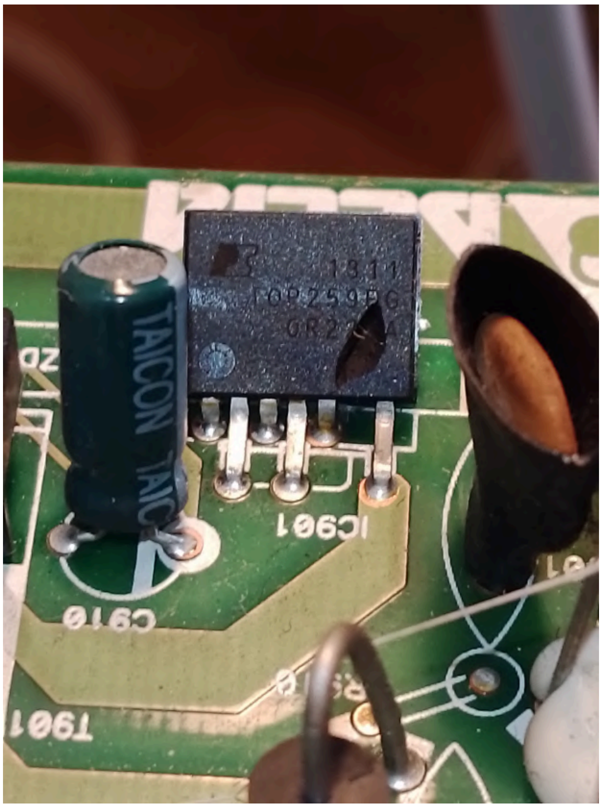

Figure 45

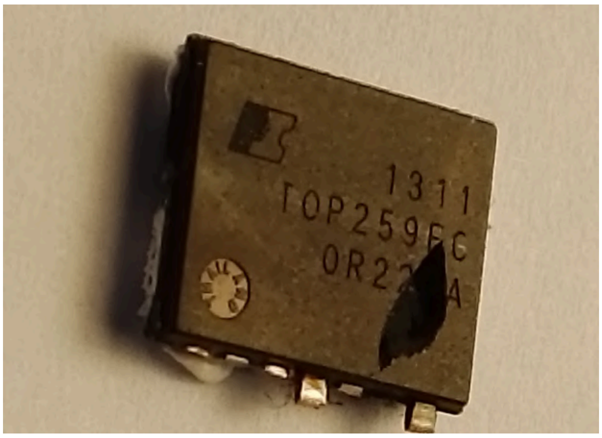

Figure 46

The flyback transfromer dimensions:

- L x l x H = 34 mm x 28.2 mm x 11.2 mm

It apparently a E28/17/11 core (EE configuration)

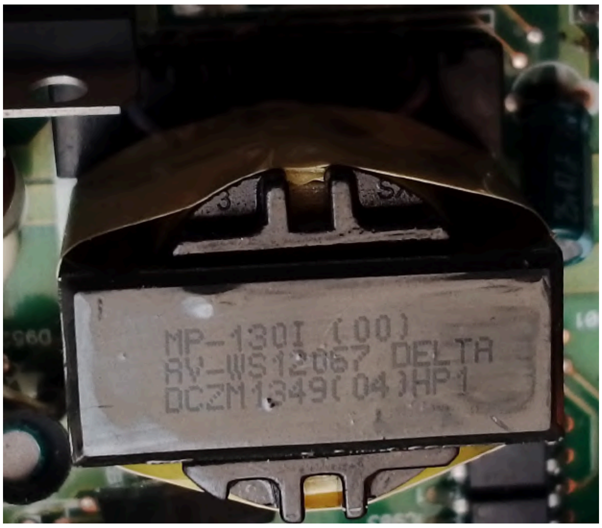

Figure 47 Trans marking

Figure 48 Overview of the transformer

This power supply use 4 optocoppler (EL816)

The PFC side is powered with an aux of the flyback with a simple regulator LM317T with R1 (2kΩ marking 2001) and R2 (22.1kΩ marking 2212), R2 has a decoupling parallel capacitor, this cofiguration generate 15V to power the controller of the PFC.